

Java Sea regional model validation

Field	Entry
Region	Indonesian Sea, LME_038. The source Java Sea domain is subregional. Regional workbook
Catch source	Sea Around Us v 50.1,1950–2019 original catch. Reference:2019 landings; all 181 exact labels, including zero catch and residual categories. Original catch
Selected article	Buchary, E.A.(1999). Evaluating the effect of the 1980 trawl ban in the Java Sea, Indonesia: An ecosystem-based approach. MSc thesis, University of British Columbia. Local main PDF verified; inherited BUCHARY-1991 folder is an alias. Original thesis Source review
Other known articles	Zainuri & Endrawati(1999):115 ha Awur Bay, a local 12-component model. Nurhakim(2003):27-group 1979 north Central Java survey model; printed diet defects prevent strict admission. Both are subregional alternatives. Awur Bay source North Central Java source
Selected model	38_38003_Java_Sea_normalized_BA_completed_(mid1970s). Static mid 1970s (1974–1976 catch context);28 source groups plus one computational import. Reported Java Sea area 471000 km ² . Accepted selected model Variant provenance
Other models from the same article	The thesis contains one documented mid 1970s static baseline and Ecosim trawl-ban simulations. A second complete, independently verified static parameter/diet model was not identified; dynamic scenarios are not separate static baselines.
Selection rationale	Article — ? Model — User-selected best available Java Sea derived variant. Full-LME and multi-decadal transfer remain provisional; source fidelity, warnings and applicability are assessed separately.
Model extraction	Source Tables 3.1,3.9,3.10 and parameter prose reviewed. Accepted Macrozoobenthos diet 0.660→1 proportional normalization and 28 signed computational BA completions are retained. BA is not measured biomass change. Canonical bytes, loaded biological inputs and source catches remain unchanged. Source/derived differences Loaded-state proof
SPPR calculation	[MANUAL — Researcher: calculation choices and reasons] [Excluded groups, scope and relevant values; other adjustments]

Regional review completed

Field	Entry
GE diagnostics	No negative SPPR entries across the basal-source columns. OK retained. rho_living=0.40177283; b=0.24546092; Detritus(28) SPPR=1.31494772.
TE diagnostics	No negative SPPR entries across the basal-source columns. WARN retained. rho_living=0.62129934; b=0.00000000; Detritus(28) SPPR=0.77389867. Marine mammals have near-zero transfer efficiency; finite results remain provisional.
Geographic fit	A. Target fraction covered by study: approximately 19–22%. B. Study fraction within target: approximately 99–100%. Low geographic confidence: approximate prose-bound reconstruction with two eastern scenarios; no explicit source study polygon.
Temporal fit	Static mid 1970s coefficients and fixed source-catch/donor weights are applied across 1950–2019 whole-LME landings, catch and discards. Reference 2019 is about 44 years later. The source guild populations and historical trawl bycatch differ from modern recorded fisheries.
Other	With Egestion is OK under accepted thresholds. All 348 entries in the three direct 29×4 source matrices are finite/nonnegative, including unfished groups. GE/Egestion threshold grades do not establish ecological validity. Historical method-specific failures/warnings remain. Mixed bony/cartilage source pools and assumed composition limit coarse labels. Full matrices and warnings
Open issues and next action	[Researcher: open issues and next actions]
Review and reproducibility	Researcher name: [name] review date: [date]

Taxon mapping and coverage

Reference:2019 landings; all exact labels, including zero catch and residual categories; no group filter.

28 source groups(+1 computational import);181 catch taxa.

Simple-chain PPR totals 177,164,712.36 t C in 2019.

39 labels lack a classic coefficient; all have zero 2019 catch and zero annual contribution. Unavailable TL/coefficient stays ?, with no invented value.

Method: protected independent simple trophic-chain coefficients; wet coefficient×catch/9 once.

[All 181 taxon mappings](#) and [descriptive Sources sheet](#)

Overall confidence	Taxa	Catch percentage	PPR percentage
High	56	33.9189%	29.0730%
Medium	75	32.9581%	48.5117%
Low	14	2.1288%	2.7273%
Very low	36	30.9943%	19.6880%
Unresolved	0	0.0000%	0.0000%

The summary uses the weaker required membership or allocation confidence. The following tables show each component separately.

Group assignment rules

Plain-language rule	Confidence	PPR percentage
Documented source taxon/family criterion supports the particular taxon.	High	58.2144%
Assumed complete source-containing family or stage set.	Medium	11.5475%
Closest represented analogue with material habitat/taxonomic mismatches.	Very low	11.0682%
Supported ecological extension to represented source members.	Medium	10.3928%
Broad reporting category approximated across compatible pools.	Very low	8.6198%
Verified historical name or spelling bridge.	High	0.1520%
Partial or competing source ecological/size fit.	Low	0.0053%

Membership evidence: [source Table3.1](#), [habitat/size rule](#), [native FAO support](#) and [all decisions](#)

Allocation weight rules

Plain-language rule	Confidence	PPR percentage
Complete printed source Harvest proportions, including genuine zeros.	Medium	48.8099%
One reviewed source group receives 100%; no numerical split.	High	48.4430%
Historical identified-catch donor proportions; population/basis mismatch.	Low	2.7471%

Percentages are each rule's share of simple-chain PPR. Complete Harvest includes genuine zero stages. Juvenile shrimp 0.008 and juvenile large demersals 0.066 are author assumptions. Living bottom structure catch is 10×SAF assumed trawl bycatch and supplies 71.43% of the generic-invertebrate allocation. Retained donor fractions use all eligible identified direct/stage donors, with no same-family restriction; mismatch remains Low. Compatible measured stage-mass partitions were not located. [Allocation evidence](#)

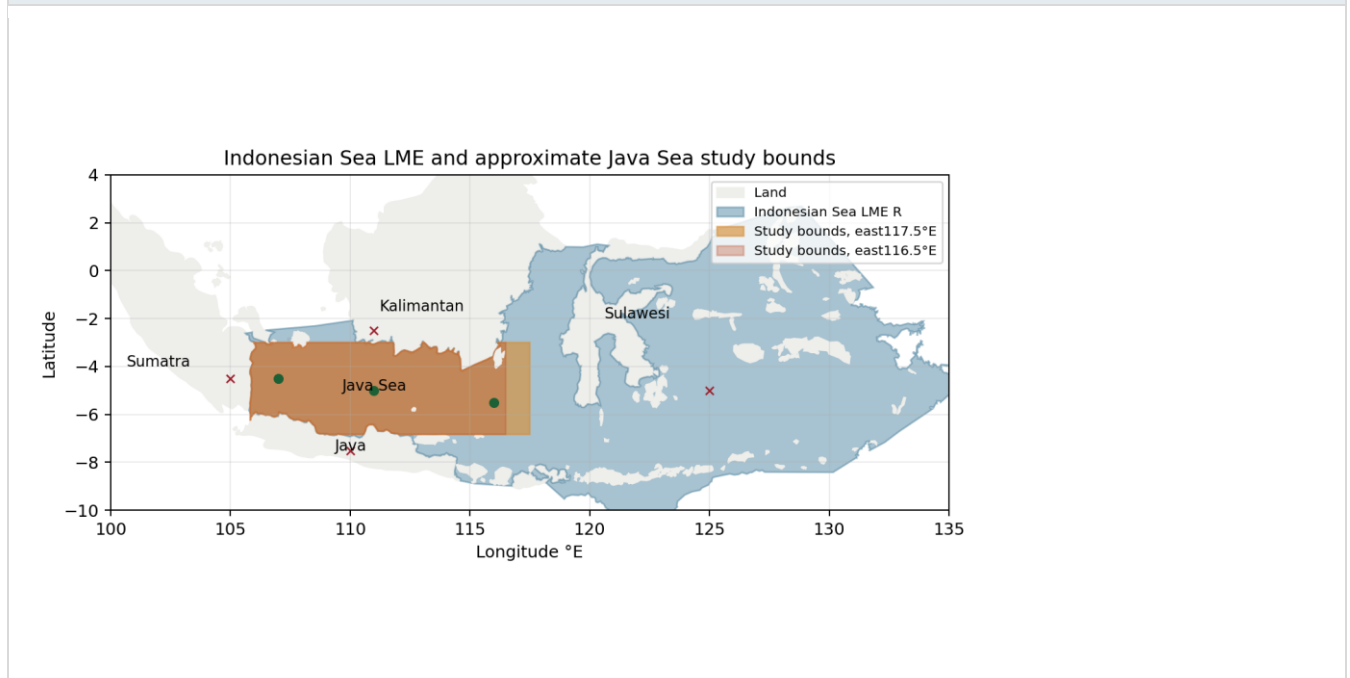
Unresolved: none. Very low: 36 labels contribute 30.9943% of landings and 19.6880% of simple-chain PPR. Broad bony fish/invertebrate pools and unlisted billfish, sharks, tarpon, catfish, flatfish and deep-water taxa use explicit approximations. Mixed-source pools retain non-target members. Very low all-source model PPR shares: GE 33.35%; TE 49.58%; With Egestion 32.20%. Broad marine fish alone supplies 30.07% of TE; Lates calcarifer analogy 18.58%. [Method-specific exposure](#)

Geographic evidence screenshots

R is the full Sea Around Us Indonesian Sea LME. S reconstructs the thesis's approximate 3°S–6°50'S, 105°50'E–116°30'/117°30'E prose bounds, removing land consistently. Figure 2.1 supplies bathymetry and provincial context; it has no explicit study polygon. A/B are approximate regional-extent estimates.

Region area

Indonesian Sea target and two source-bound scenarios



Target R: 2,267,636 km². Reconstructed water areas: 441,114/488,277 km²; intersections: 439,052/486,216 km². A ≈ 19–22%, B ≈ 99–100%. Source-reported 471,000 km² is separate context. All seven stated water/land controls pass. [Target polygon](#) | [Approximate bounds](#)

Study area in the article

Original source Figure 2.1: Java Sea geography and bathymetry

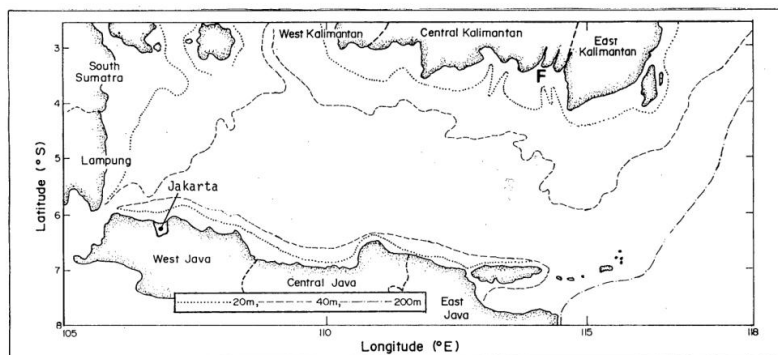


Figure 2.1 Map of the Java Sea showing its bathymetry, and the bordering eight provinces and one municipality (adapted from Martosubroto 1996).

Original PDF 30/printed 21, rotated for readability. Its graticule and bordering provinces inform the reconstruction; source latitude bounds can truncate local coast-following water. Natural Earth 1:50 million land removal is generalized. Two eastern bounds express source ambiguity, not statistical confidence intervals. No geographic multiplier is applied. [Original figure](#) | [Areas and limits](#)

Original Table 3.1, PDF 58/printed 49: source biological groups and representative taxa. The family summary does not exhaust modern taxonomic membership.

Table 3.1 Species composition for the 28 functional groups of the Java Sea model ^a

No.	Functional Group	Species included
1.	Benthic producers	Marine algae (epilithic algae, endolithic algae, reef turf algae, benthic fleshy algae, macroalgae, and benthic algae) and spermatophytes (sea grass)
2.	Phytoplankton	Dominated by diatoms (<i>Bacteriastrum</i> sp., <i>Biddulphia</i> sp., <i>Bacillaria</i> sp., <i>Chaetoceros</i> sp., <i>Coscinodiscus</i> sp., <i>Ditylum</i> sp., <i>Eucampia</i> sp., <i>Hemiaulus</i> sp., <i>Hemidiscus</i> sp., <i>Lauderia</i> sp., <i>Leptocylindricus</i> sp., <i>Nitzschia</i> sp., <i>Rhizosolenia</i> sp., <i>Skeletonema</i> sp., <i>Streptotheca</i> sp., <i>Thalassiothrix</i> sp./ <i>Thalassionema</i> sp.) and dinoflagellates (<i>Ceratium</i> sp., <i>Dinophysis</i> sp., and <i>Peridinium</i> sp.)
3.	Small herbivorous zooplankton	Copepodes, ostracodes, bivalve larvae, cirripedia larvae, cladocerans, echinoderm larvae, larvacea, other mollusk larvae, and larvacea
4.	Large herbivorous zooplankton	Mysids, sergestids, euphausiids, amphipodes, luciferidae, and other decapod larvae
5.	Carnivorous zooplankton	Chaetognaths (<i>Sagitta</i> spp.), annelids and ichthyoplankton
6.	Jellyfish	All medusoid form of cnidarians (hydrozoa and scyphozoa only); however, scyphozoan medusae dominates
7.	Benthic infauna	Includes all burrowing benthos of the size ≤ 1.0 mm, viz., polychaetes, mollusks, echinoderms, crustaceans, sipunculans, and benthic stage larvae of other larger organisms
8.	Structure-associated fish (SAF)	This group is designed to include all biota (notably fish) that ecologically depend (directly and/or indirectly) on living bottom structure (LBS, see components of group #10) to complete their life cycle. For this model, it includes fish of the families Balistidae, demersal Carangidae, Lethrinidae, Lutjanidae, Pentapodidae, Platacidae, Polynemidae, Sparidae, Sphyrnidae, Scaridae, Acanthuridae, and Tetraodontidae
9.	Macrozoobenthos	This group includes all larger (size > 1.0 mm) mollusks and echinoderms, such as conch, oyster, scallops, clams, cockles, mussels, sea urchins, sea cucumbers, and sea stars
10.	Living bottom structure (LBS)	This functional group comprised of all biogenic organisms that live on and/or are attached to the sea floor, such as sponges (notably giant cup sponges, viz. <i>Poterion nautilus</i> , <i>P. neptuni</i> and <i>P. amphitrite</i>), gorgonians (sea fans and sea whips), soft corals, sea pens, sea anemones, etc.
11.	Juvenile penaeid shrimps	This group includes all juvenile shrimps, not just penaeid shrimps. However, the parameterization of this group is based on the juveniles of <i>Penaeus</i> spp. and <i>Metapenaeus</i> spp.
12.	Large pelagic predators (juvenile)	Juveniles of the families Chirocentridae, large Scombridae, and Trichiuridae